

SAAD MUHAMMAD
F2023-009

Assignment 4
Section C

Analysis of
Algorithms

Q1. $A_1 \cdot A_2 \cdot A_3 \cdot A_4 \cdot A_5$
(5x4) (4x6) (6x2) (2x7) (7x3)
 $d_0 \ d_1 \ d_2 \ d_3 \ d_4 \ d_5$

$$c[i, j] = \min_{i \leq k < j} \{ c[i, k] + c[k+1, j] + d_{i-1} \times d_k \times d_j \}$$

		j →				
	C	1	2	3	4	5
i ↓	1	0	120	88	158	160
	2		0	48	104	114
	3			0	84	78
	4				0	42
	5					0

		j →				
	k	1	2	3	4	5
i ↓	1		1	18	3	3
	2			2	3	2
	3				3	3
	4					4
	5					

$$c[1, 2] = \min_{1 \leq k < 2} \{ c[1, 1] + c[2, 2] + d_0 \cdot d_1 \cdot d_2 \}$$

$$0 + 0 + (5 \cdot 4 \cdot 6) = 120 \checkmark$$

$$c[2, 3] = \min_{2 \leq k < 3} \{ c[2, 2] + c[3, 3] + d_1 \cdot d_2 \cdot d_3 \}$$

$$0 + 0 + (4 \cdot 6 \cdot 2) = 48 \checkmark$$

$$c[3, 4] = \min_{3 \leq k < 4} \{ c[3, 3] + c[4, 4] + d_2 \cdot d_3 \cdot d_4 \}$$

$$0 + 0 + (6 \cdot 2 \cdot 7) = 84 \checkmark$$

$$c[4, 5] = \min_{4 \leq k < 5} \{ c[4, 4] + c[5, 5] + d_3 \cdot d_4 \cdot d_5 \}$$

$$0 + 0 + (2 \cdot 7 \cdot 3) = 42 \checkmark$$

$$c[1, 3] = \min_{k=1} \left\{ \begin{aligned} &c[1, 1] + c[2, 3] + d_0 \cdot d_1 \cdot d_3 \\ &k=2 \left\{ \begin{aligned} &c[1, 2] + c[3, 3] + d_1 \cdot d_2 \cdot d_3 \\ &\rightarrow 0 + 48 + (5 \cdot 4 \cdot 2) = 88 \checkmark \\ &\rightarrow 120 + 0 + (4 \cdot 6 \cdot 2) = 168 \end{aligned} \right. \end{aligned} \right.$$

$$c[2,4] = \min_{2 \leq k < 4} \left\{ \begin{array}{l} k=2 \left\{ c[2,2] + c[3,4] + d_2 \cdot d_2 \cdot d_4 \right\} \\ k=3 \left\{ c[2,3] + c[4,4] + d_1 \cdot d_3 \cdot d_4 \right\} \end{array} \right\}$$

$\rightarrow 0 + 84 + (4 \cdot 6 \cdot 7) = 252$
 $\rightarrow 48 + 0 + (4 \cdot 2 \cdot 7) = 104 \checkmark$

$$c[3,5] = \min_{3 \leq k < 5} \left\{ \begin{array}{l} k=3 \left\{ c[3,3] + c[4,5] + d_2 \cdot d_3 \cdot d_5 \right\} \\ k=4 \left\{ c[3,4] + c[5,5] + d_2 \cdot d_4 \cdot d_5 \right\} \end{array} \right\}$$

$\rightarrow 0 + 42 + (6 \cdot 2 \cdot 3) = 78 \checkmark$
 $\rightarrow 84 + 0 + (6 \cdot 7 \cdot 3) = 210$

$$c[1,4] = \min_{1 \leq k < 4} \left\{ \begin{array}{l} k=1 \left\{ c[1,1] + c[2,4] + d_0 + d_1 + d_4 \right\} \\ k=2 \left\{ c[1,2] + c[3,4] + d_0 + d_2 + d_4 \right\} \\ k=3 \left\{ c[1,3] + c[4,4] + d_0 + d_3 + d_4 \right\} \end{array} \right\}$$

$\rightarrow 0 + 104 + (5 \cdot 4 \cdot 7) = 244$
 $\rightarrow 120 + 84 + (5 \cdot 6 \cdot 7) = 414$
 $\rightarrow 88 + 0 + (5 \cdot 2 \cdot 7) = 158 \checkmark$

$$c[2,5] = \min_{2 \leq k < 5} \left\{ \begin{array}{l} k=2 \left\{ c[2,2] + c[3,5] + d_1 \cdot d_2 \cdot d_5 \right\} \\ k=3 \left\{ c[2,3] + c[4,5] + d_1 \cdot d_3 \cdot d_5 \right\} \\ k=4 \left\{ c[2,4] + c[5,5] + d_1 \cdot d_4 \cdot d_5 \right\} \end{array} \right\}$$

$\rightarrow 0 + 78 + (4 \cdot 6 \cdot 3) = 150$
 $\rightarrow 48 + 42 + (4 \cdot 2 \cdot 3) = 114 \checkmark$
 $\rightarrow 104 + 0 + (4 \cdot 7 \cdot 3) = 188$

$$c[1,5] = \min_{1 \leq k < 5} \left\{ \begin{array}{l} k=1 \left\{ c[1,1] + c[2,5] + d_0 + d_1 + d_5 \right\} \Rightarrow 0 + 114 + (5 \cdot 4 \cdot 3) = 174 \\ k=2 \left\{ c[1,2] + c[3,5] + d_0 + d_2 + d_5 \right\} \Rightarrow 120 + 78 + (5 \cdot 6 \cdot 3) = 288 \\ k=3 \left\{ c[1,3] + c[4,5] + d_0 + d_3 + d_5 \right\} \Rightarrow 88 + 42 + (5 \cdot 2 \cdot 3) = 160 \checkmark \\ k=4 \left\{ c[1,4] + c[5,5] + d_0 + d_4 + d_5 \right\} \Rightarrow 158 + 0 + (5 \cdot 7 \cdot 3) = 263 \end{array} \right\}$$

$\Rightarrow$

Q2.

Activity	Start	Finish
A1	1	2
A2	3	4
A3	0	6
A4	5	7
A5	8	9
A6	5	9

Greedy Method

- Already sorted according to finish time

① Select A1

Finish = 2 {A1}

② $3 \geq 2 \rightarrow$ Select

Finish = 4 {A1, A2}

③ $0 < 4 \rightarrow$ Reject

Finish = X

④ $5 \geq 4 \rightarrow$ Select

Finish = 7 {A1, A2, A4}

⑤ $8 \geq 7 \rightarrow$ Select

Finish = 9 {A1, A2, A4, A5}

⑥ $5 < 9 \rightarrow$ Reject

Finish = X

Max Activity
Max Activities = 4